R99/B | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 34 | update 48960 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.0580	66.2037	11.416971	11.427978	11.371692	-0.1923	+0.3953	0.4055
TEST ALL	71.3642	66.1802	11.373966	11.385299	11.329932	-0.1969	+0.3862	0.3945

Dataset	n	NSS Top1 (%)	R99 Top1 (%)	NSS cost	R99 cost
TSP TEST	4500	45.888889	47.333333	7.970400719	7.972188681
CVRP TEST	4500	45.977778	48.111111	18.004802368	18.010914797
TSPLIB	49	20.408163	32.653061	8.217907074	8.200639919
CVRPLIB	100	70.000000	63.000000	67.101111717	67.215689192

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 CE .35 / comparison .35 / risk .02 / R-Drop .35

R99/B | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.0580	66.2037	11.416971	11.427978	11.371692	-0.1923	+0.3953	0.4055	per-problem TRAIN SBS
TSP	46.7778	33.1111	8.025487	8.045354	7.990533	-0.2469	+0.4374	0.4278	T2T
CVRP	46.1778	18.7111	18.162288	18.269377	18.064087	-0.5862	+0.5436	0.7253	RELD_CVRP
ATSP	77.0222	47.4889	1.638603	1.676905	1.632471	-2.2841	+0.3756	0.3714	UNICO_MatPOENet
OVRP	41.4889	35.7556	7.461453	7.469579	7.415777	-0.1088	+0.6159	0.6277	MoSES_RF
VRPB	81.0222	80.8444	9.356186	9.354876	9.318750	+0.0140	+0.4017	0.4051	reld_mt
VRPL	38.0222	33.0889	12.431215	12.437959	12.383444	-0.0542	+0.3858	0.3943	MoSES_RF
VRPTW	80.0444	79.9556	18.818812	18.819431	18.766139	-0.0033	+0.2807	0.2963	reld_mt
OVRPTW	83.8000	83.7778	10.143147	10.143730	10.116859	-0.0057	+0.2598	0.2574	reld_mt
OVRPB	88.4667	88.5111	6.357295	6.357254	6.341078	+0.0006	+0.2557	0.2656	reld_mt
OVRPL	41.2000	35.9111	7.437374	7.445709	7.390416	-0.1120	+0.6354	0.6529	MoSES_RF
VRPBL	78.7111	78.5111	9.368392	9.371523	9.328366	-0.0334	+0.4291	0.4319	reld_mt
VRPBTW	78.4667	78.4222	19.947726	19.950430	19.851084	-0.0136	+0.4868	0.4640	reld_mt
VRPLTW	78.8000	78.7556	18.765547	18.765871	18.707843	-0.0017	+0.3085	0.3249	reld_mt
OVRPBL	89.2889	89.2000	6.369390	6.370093	6.352824	-0.0110	+0.2608	0.2722	reld_mt
OVRPBTW	83.5556	83.4444	10.660571	10.661098	10.623245	-0.0049	+0.3514	0.3246	reld_mt
OVRPLTW	83.5111	83.4889	10.157404	10.157423	10.129818	-0.0002	+0.2723	0.2767	reld_mt
VRPBLTW	79.5333	79.4000	19.794113	19.797044	19.706420	-0.0148	+0.4450	0.4336	reld_mt
OVRPBLTW	83.1556	83.2889	10.610473	10.609955	10.571302	+0.0049	+0.3705	0.3477	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R99/B | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.3642	66.1802	11.373966	11.385299	11.329932	-0.1969	+0.3862	0.3945	per-problem TRAIN SBS
TSP	47.3333	34.5333	7.972189	7.989621	7.937593	-0.2182	+0.4358	0.4295	T2T
CVRP	48.1111	17.9333	18.010915	18.124983	17.918509	-0.6293	+0.5157	0.6757	RELD_CVRP
ATSP	76.9333	47.2444	1.635366	1.674207	1.629441	-2.3200	+0.3636	0.3626	UNICO_MatPOENet
OVRP	41.2000	33.6444	7.445269	7.455750	7.400487	-0.1406	+0.6051	0.6151	MoSES_RF
VRPB	80.6000	80.4000	9.349648	9.351810	9.315298	-0.0231	+0.3687	0.3716	reld_mt
VRPL	37.1111	32.2667	12.349793	12.356138	12.300958	-0.0514	+0.3970	0.4062	MoSES_RF
VRPTW	81.1333	80.8444	18.689093	18.690545	18.637408	-0.0078	+0.2773	0.2887	reld_mt
OVRPTW	83.9778	83.9333	10.092605	10.093063	10.068135	-0.0045	+0.2430	0.2338	reld_mt
OVRPB	88.0889	88.1333	6.371146	6.370826	6.352225	+0.0050	+0.2979	0.3056	reld_mt
OVRPL	42.2444	34.6000	7.463637	7.472542	7.418033	-0.1192	+0.6148	0.6265	MoSES_RF
VRPBL	79.3111	79.4222	9.395245	9.396379	9.355967	-0.0121	+0.4198	0.4177	reld_mt
VRPBTW	78.7556	78.7556	19.666195	19.665725	19.577666	+0.0024	+0.4522	0.4480	reld_mt
VRPLTW	79.2889	79.2444	18.802560	18.803136	18.745143	-0.0031	+0.3063	0.3211	reld_mt
OVRPBL	89.2444	89.2889	6.375531	6.375741	6.359310	-0.0033	+0.2551	0.2665	reld_mt
OVRPBTW	84.9333	84.8222	10.658950	10.659591	10.624748	-0.0060	+0.3219	0.2948	reld_mt
OVRPLTW	82.9111	82.8667	10.085770	10.086112	10.059001	-0.0034	+0.2661	0.2674	reld_mt
VRPBLTW	79.7333	79.7333	19.705716	19.707068	19.613284	-0.0069	+0.4713	0.4521	reld_mt
OVRPBLTW	83.6444	83.5778	10.661765	10.662149	10.625574	-0.0036	+0.3406	0.3177	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R99/B | LIB and locked provenance

Current R99 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	32.6531	10.2041	8.200640	8.240435	8.142167	-0.4829	+0.7181	0.7750
CVRPLIB	63.0000	45.0000	67.215689	67.245364	66.945093	-0.0441	+0.4042	0.5303

SHA256: ccc0ae35e450d184ec9328d7588da85a1512a3508bfa3018942c2c854f4c8dd6

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval input-only distance messages, A row-softmax or B fixed64 Sinkhorn; FP32 transform then FP16 quantization; outgoing/incoming one/two-hop raw8; Linear32to128 before original four instance layers; R88B physical TSP node0; hanc

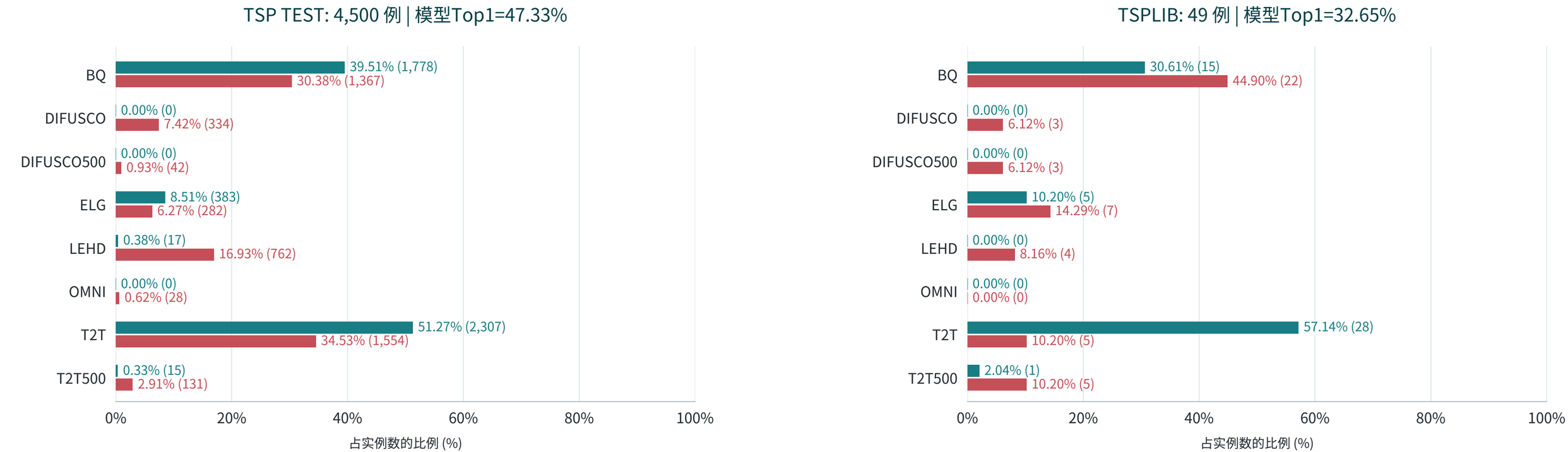
Normalization: new81000 TRAIN only; no TEST/LIB refit.

R99/B | TSP | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

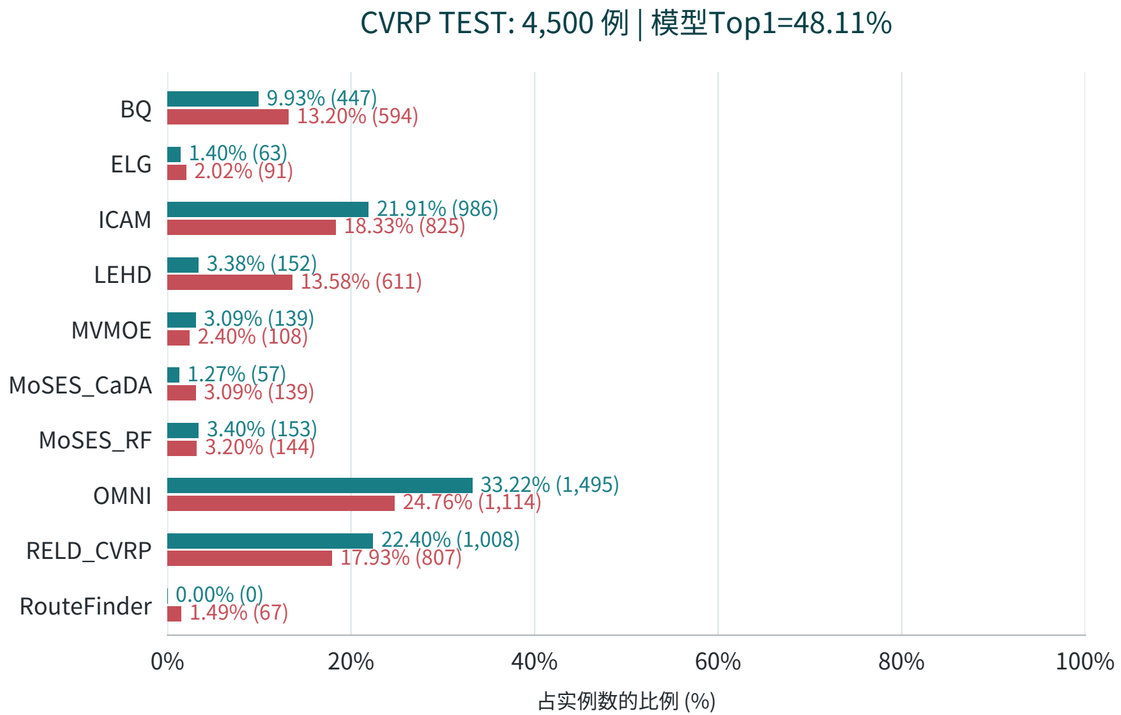
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

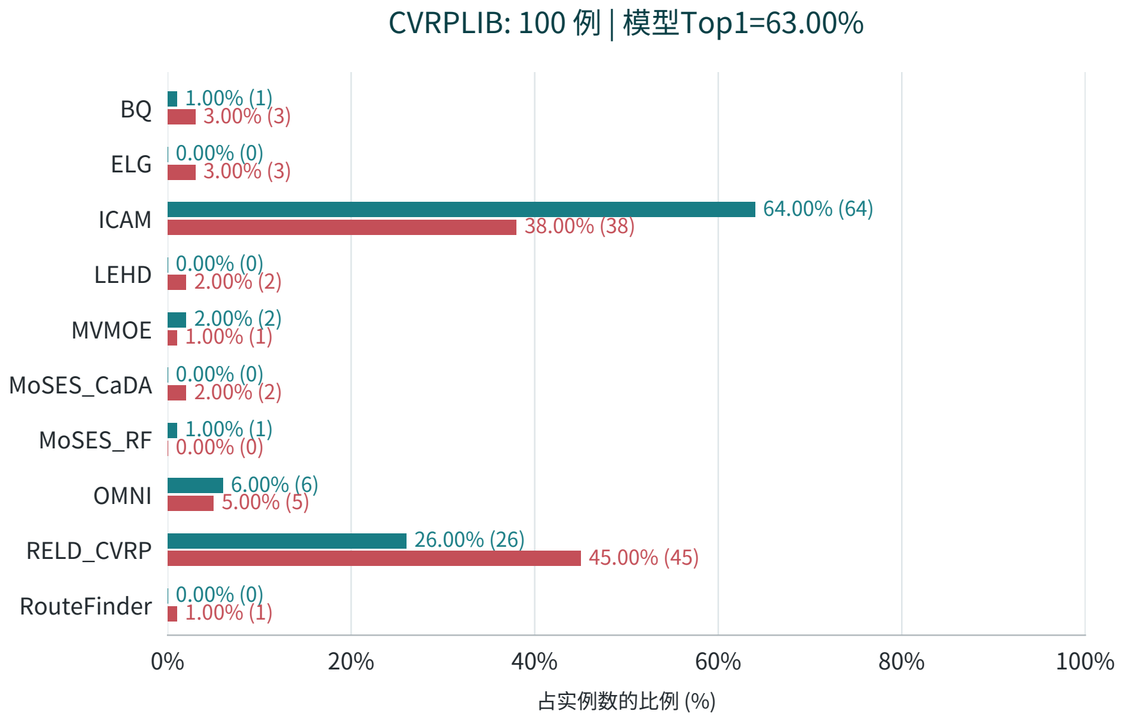
R99/B | CVRP | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

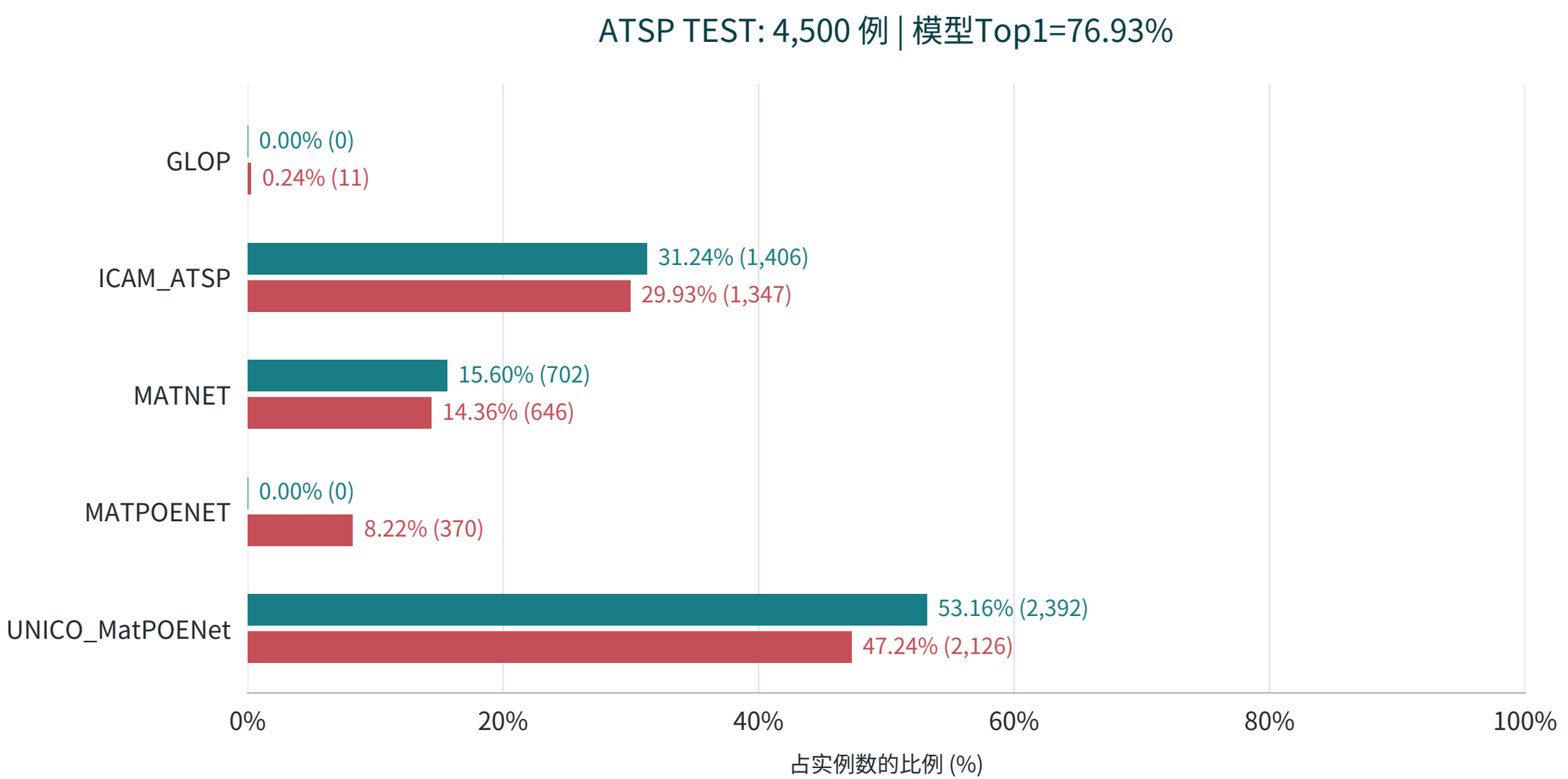
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | ATSP | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

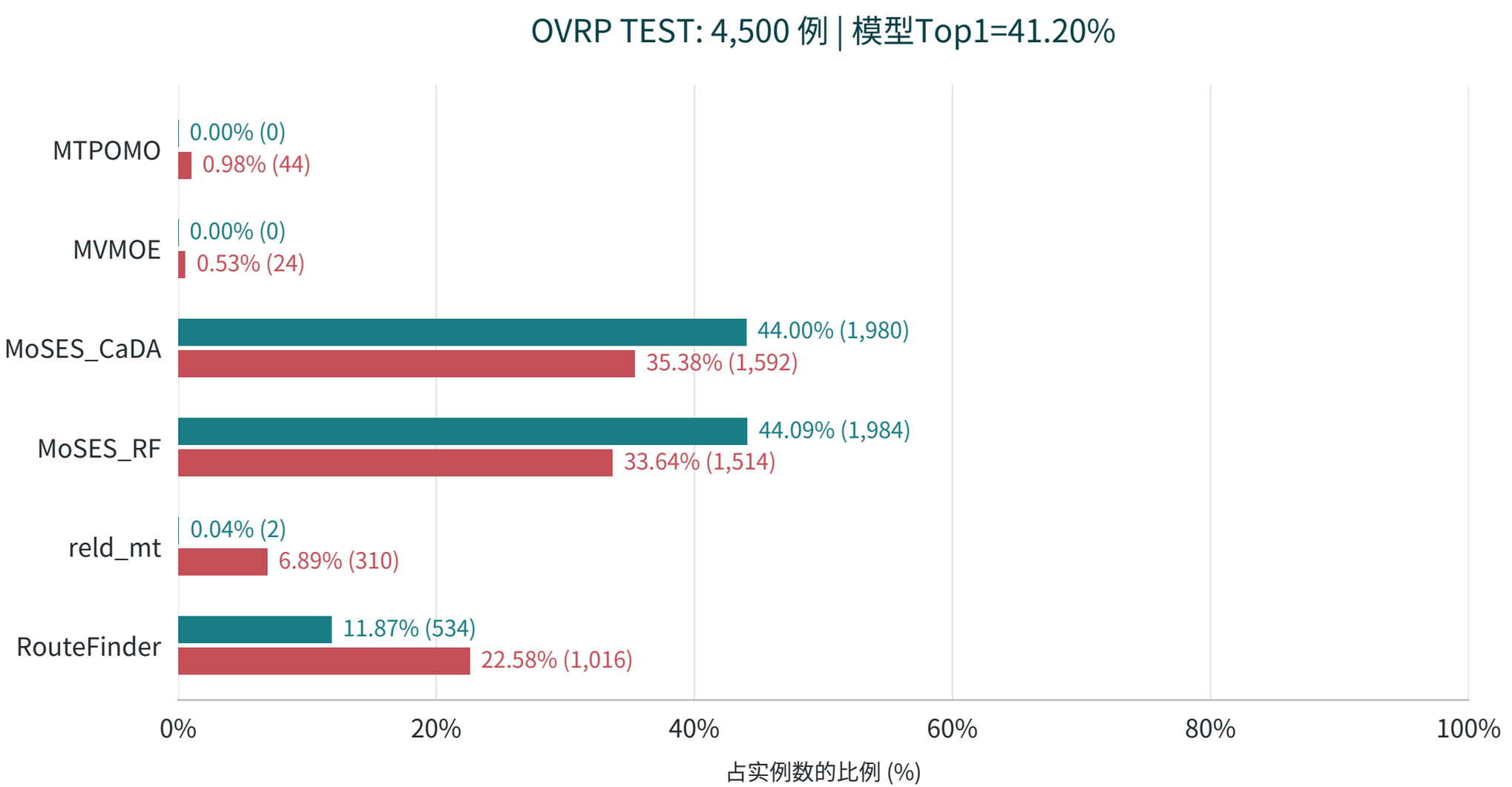
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | OVRP | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

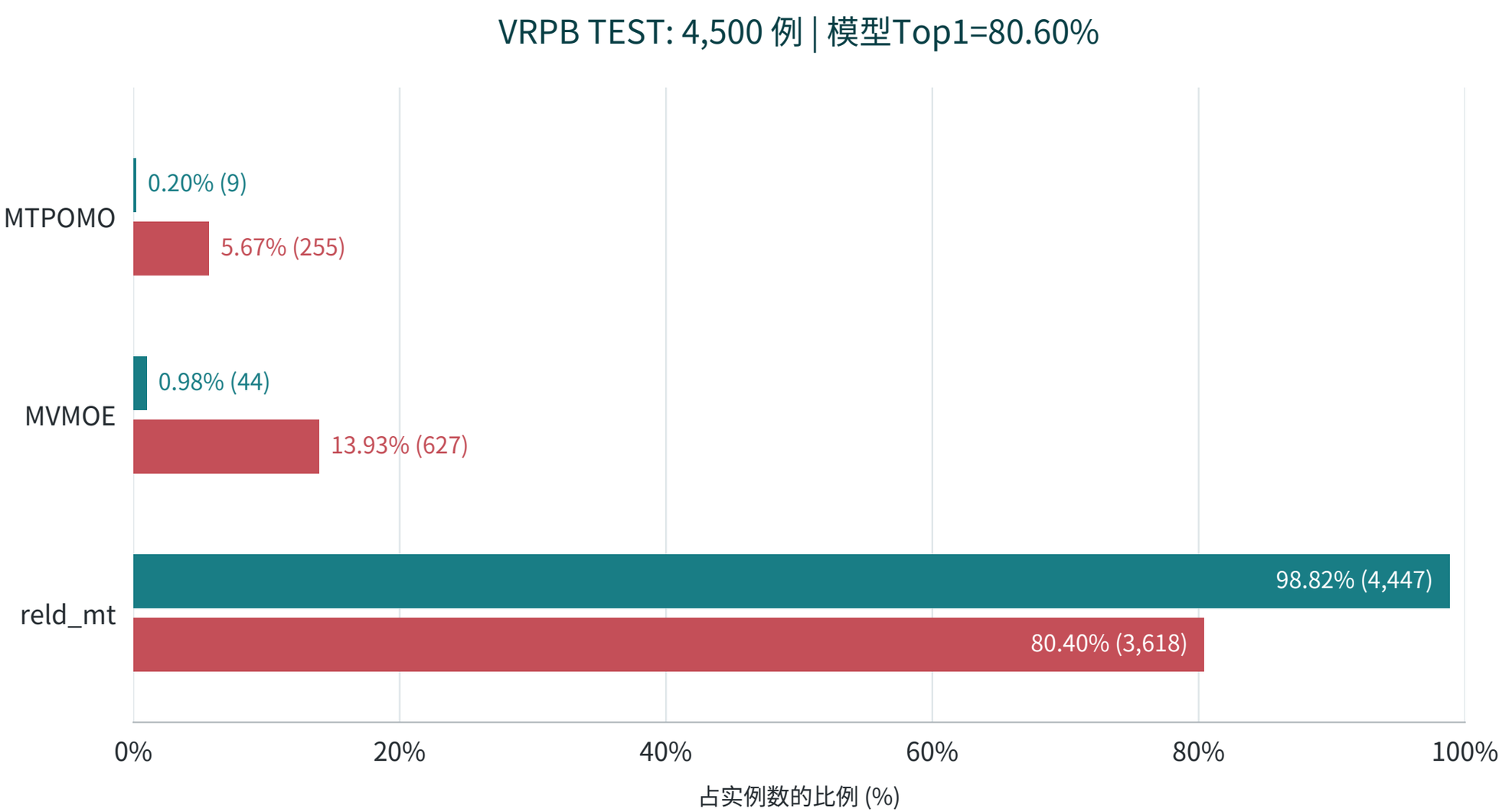
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | VRPB | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

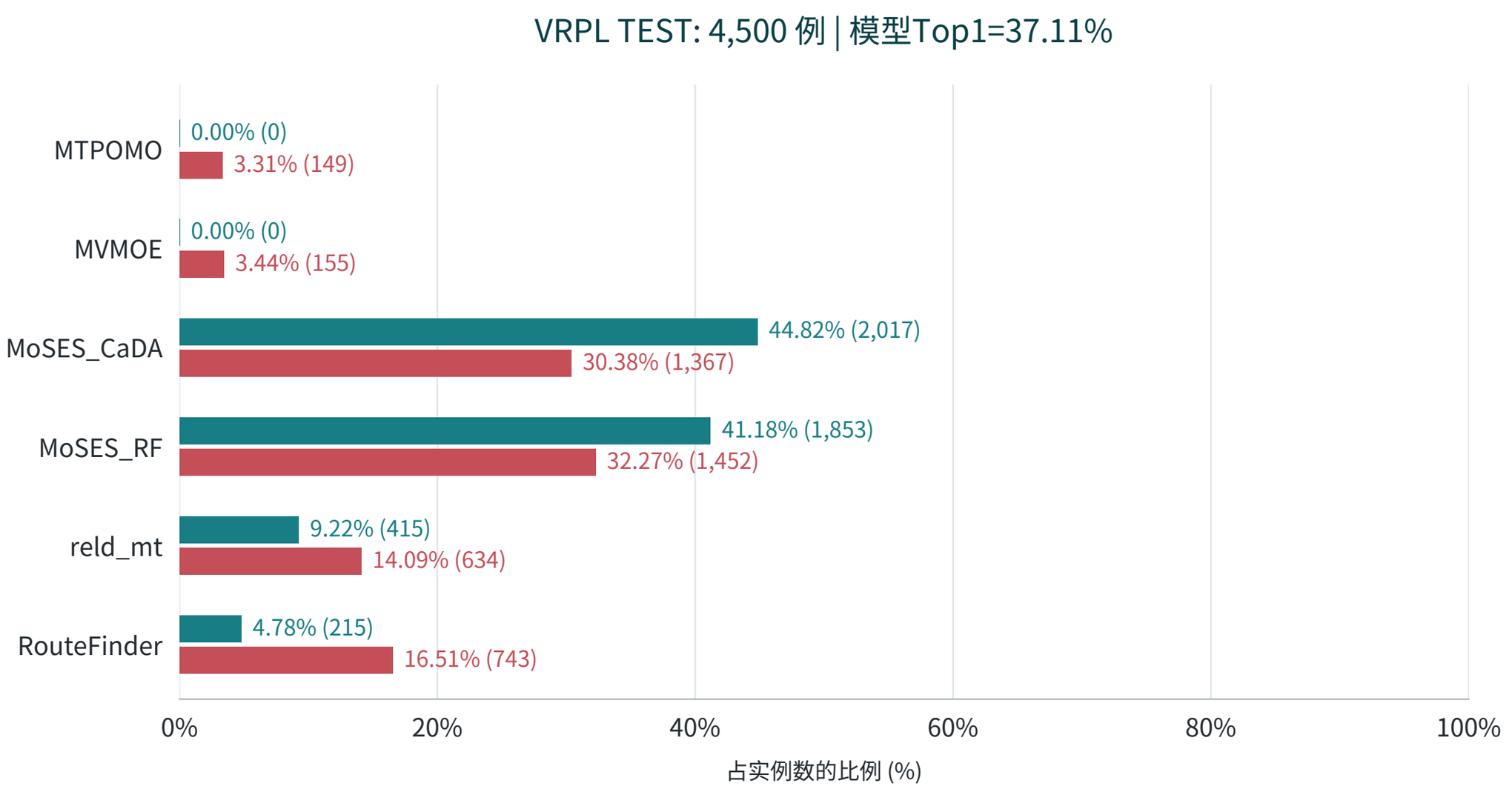
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | VRPL | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

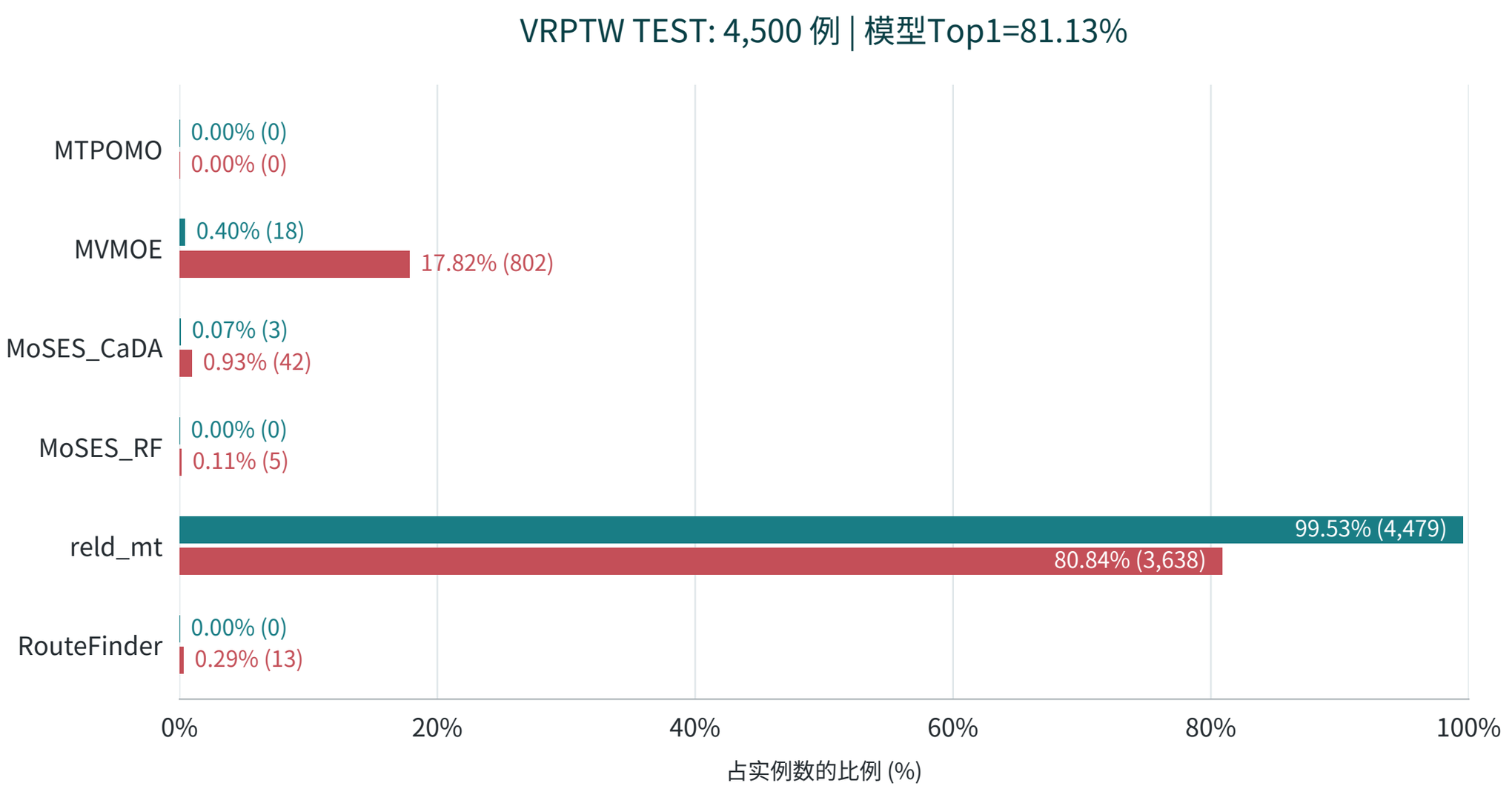
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | VRPTW | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

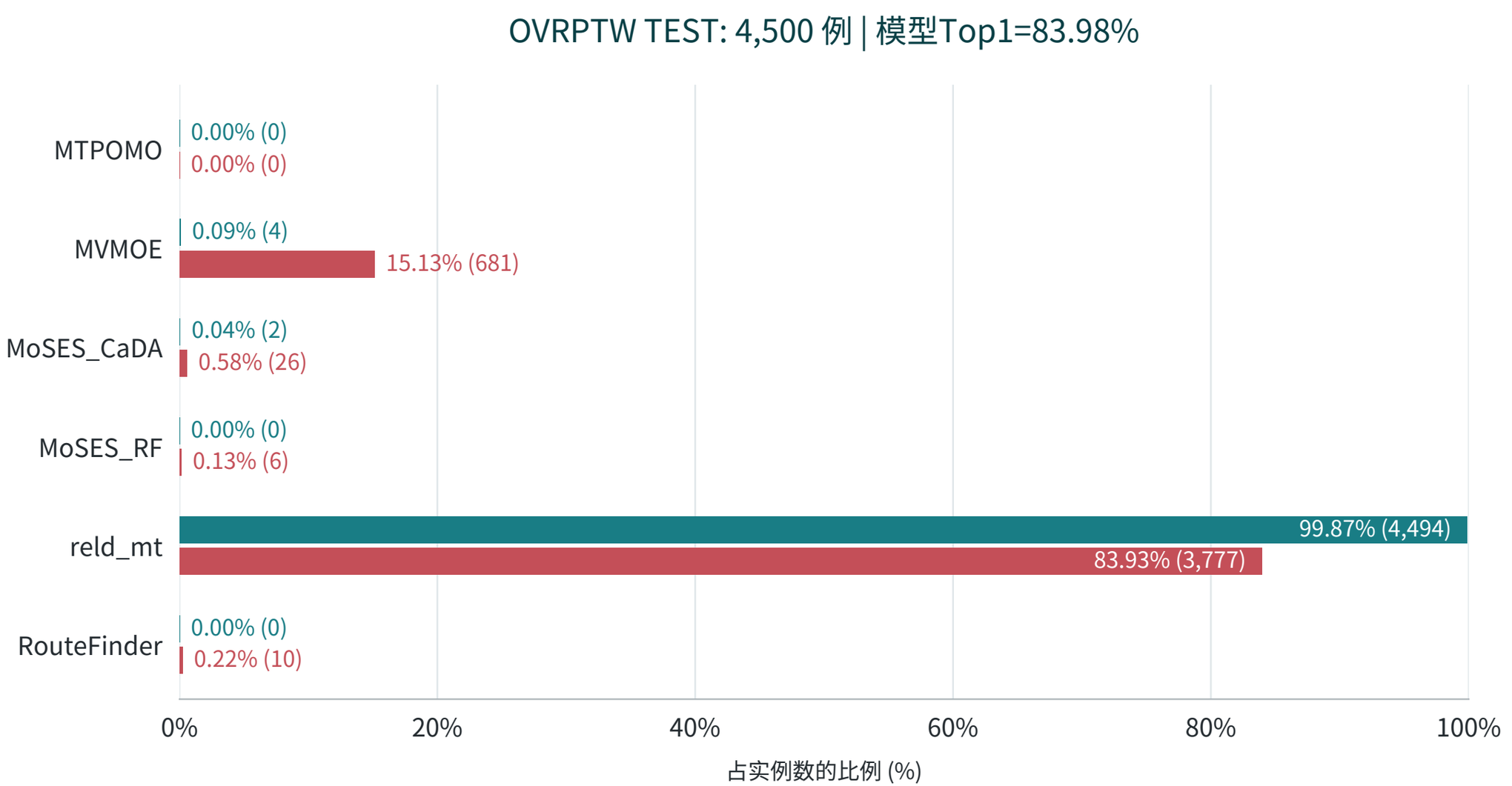
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | OVRPTW | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

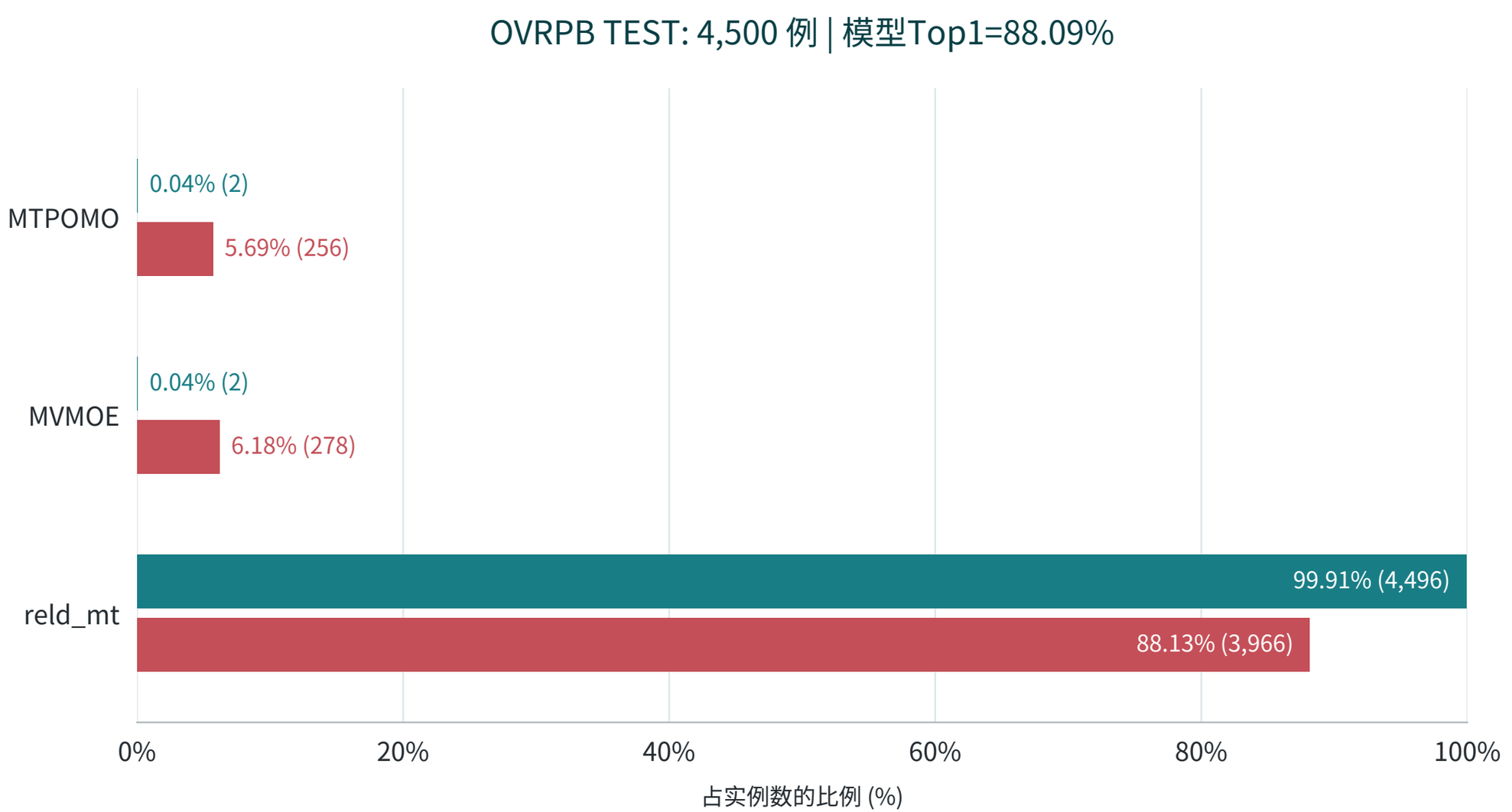
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | OVRPB | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

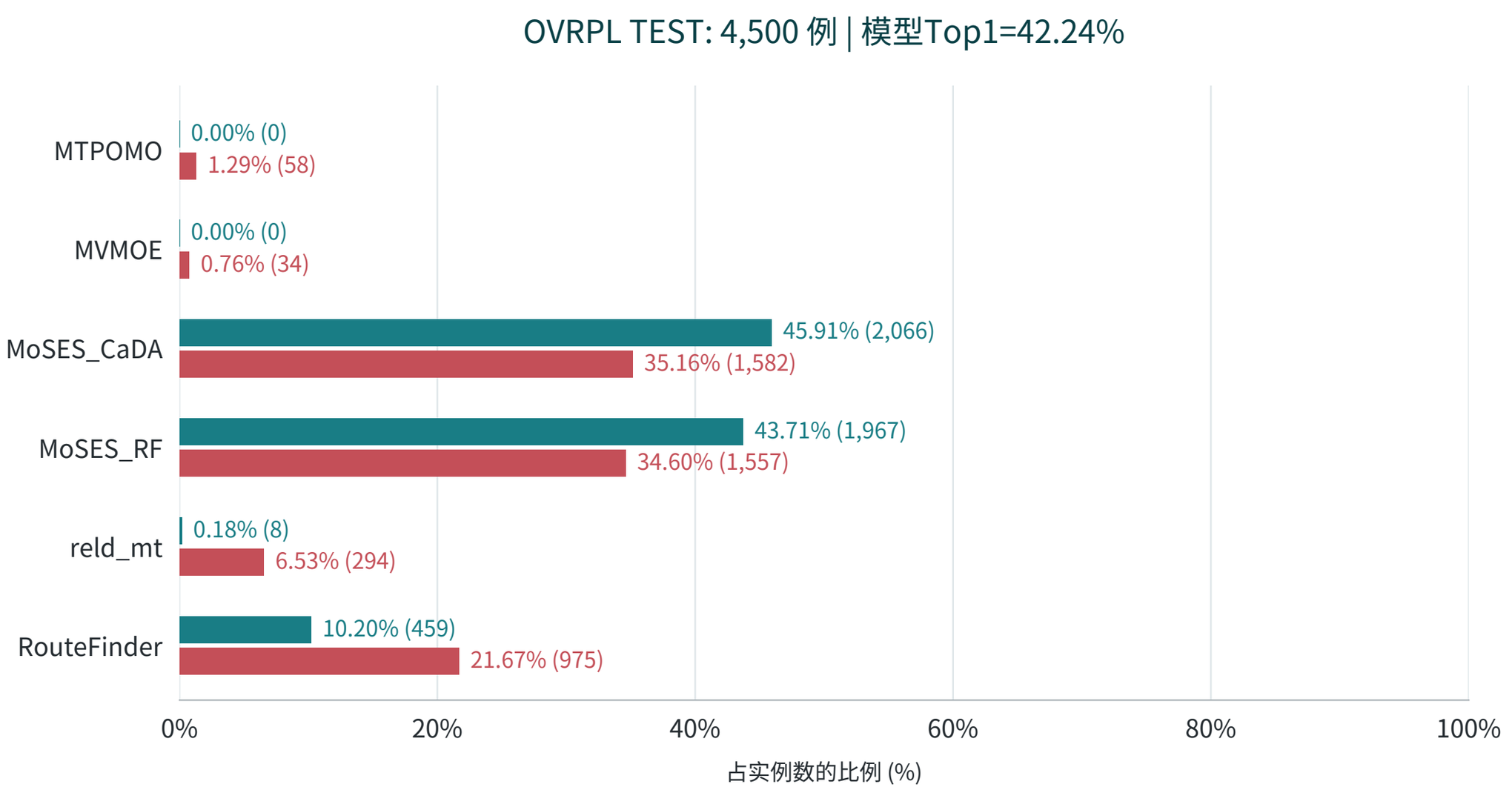
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | OVRPL | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

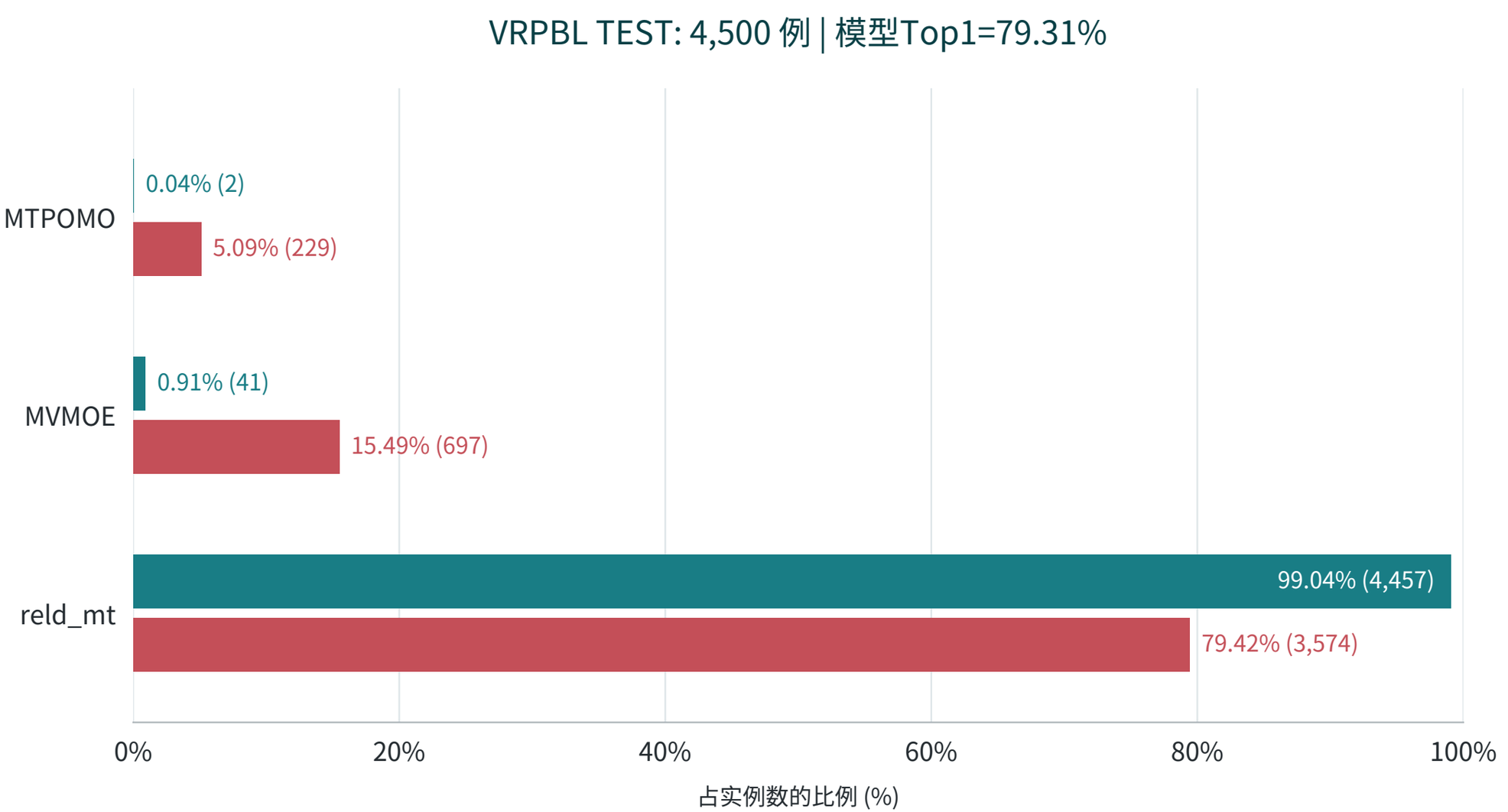
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | VRPBL | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

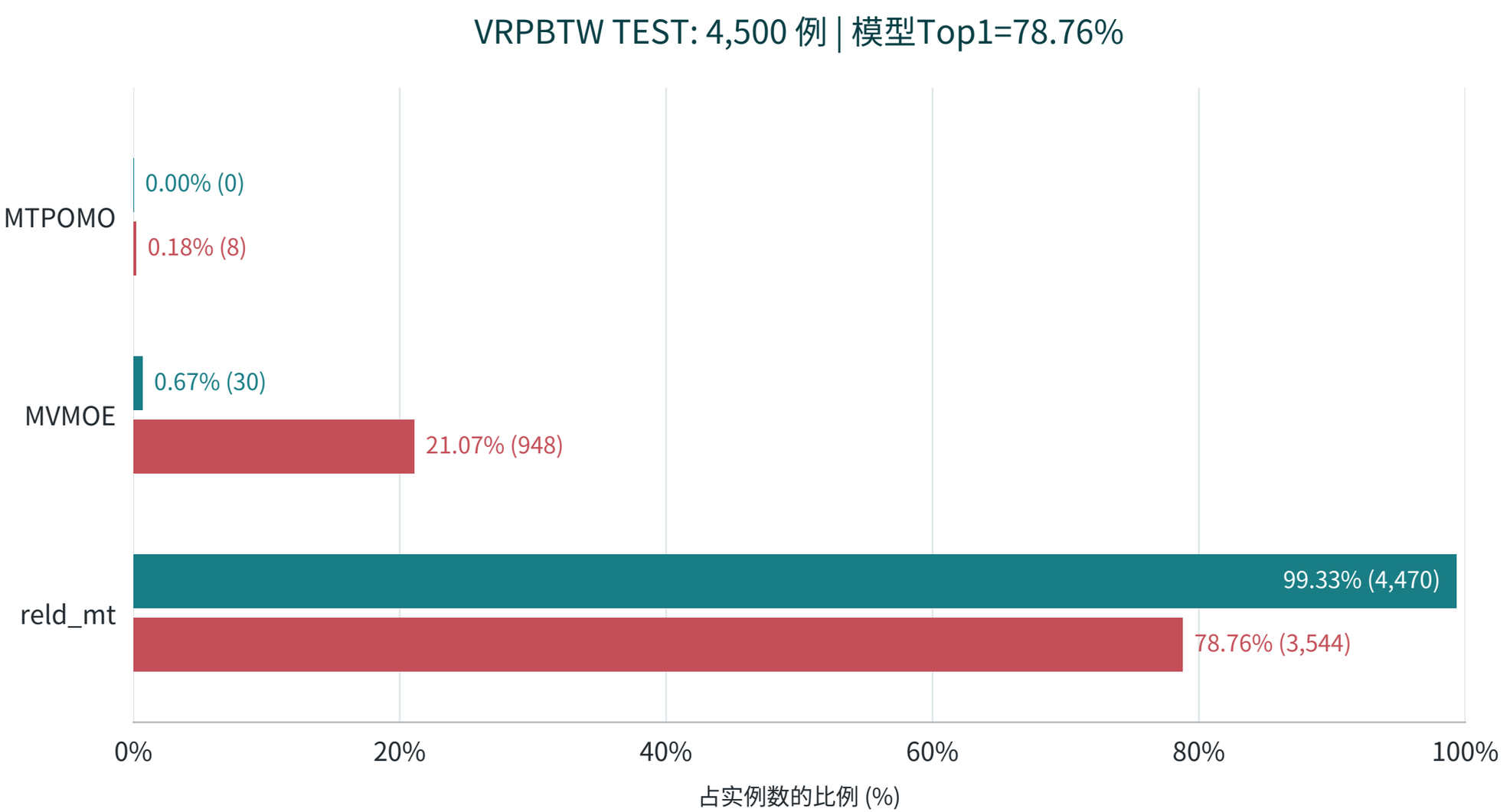
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | VRPBTW | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

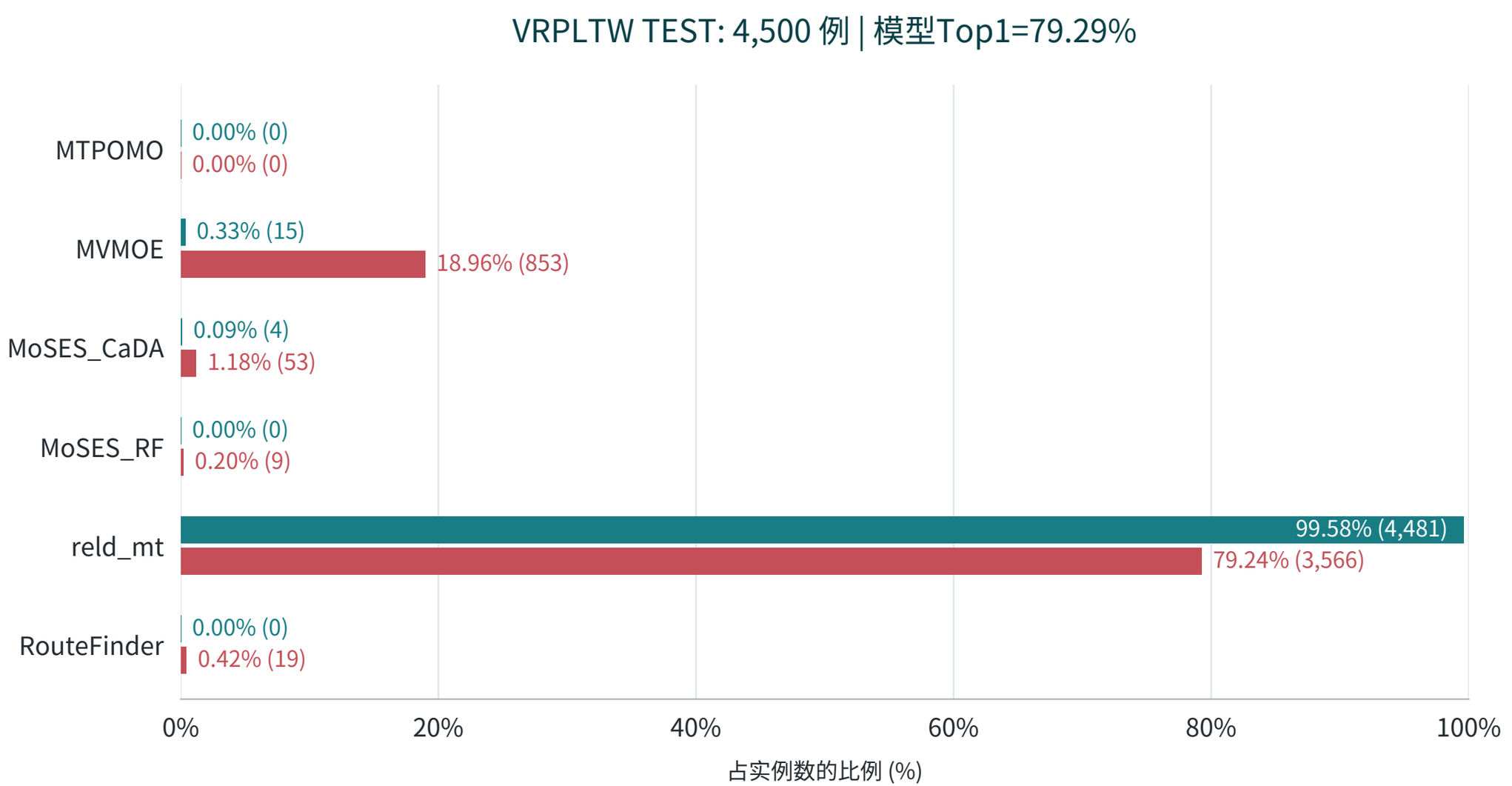
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | VRPLTW | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

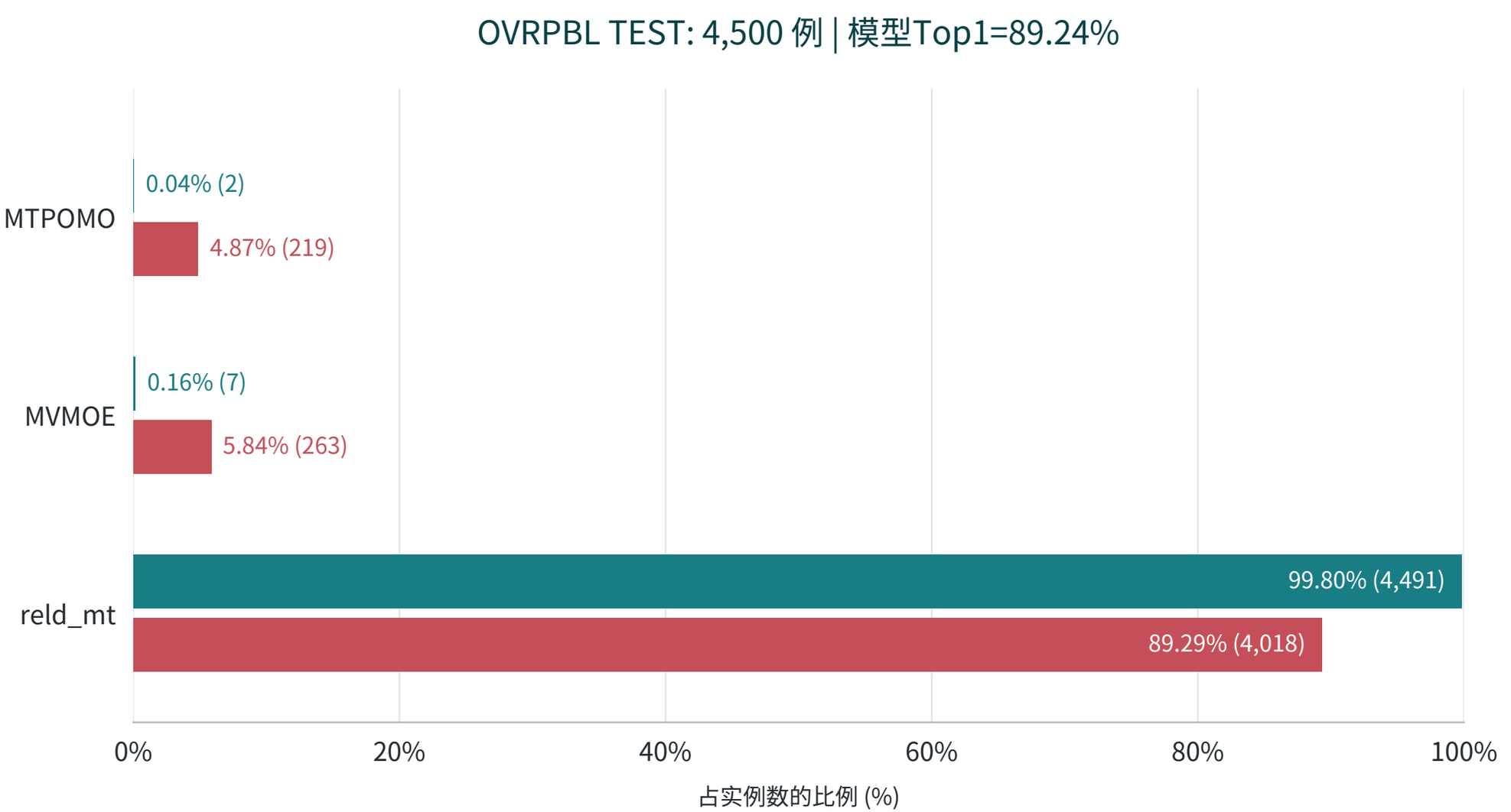
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | OVRPBL | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

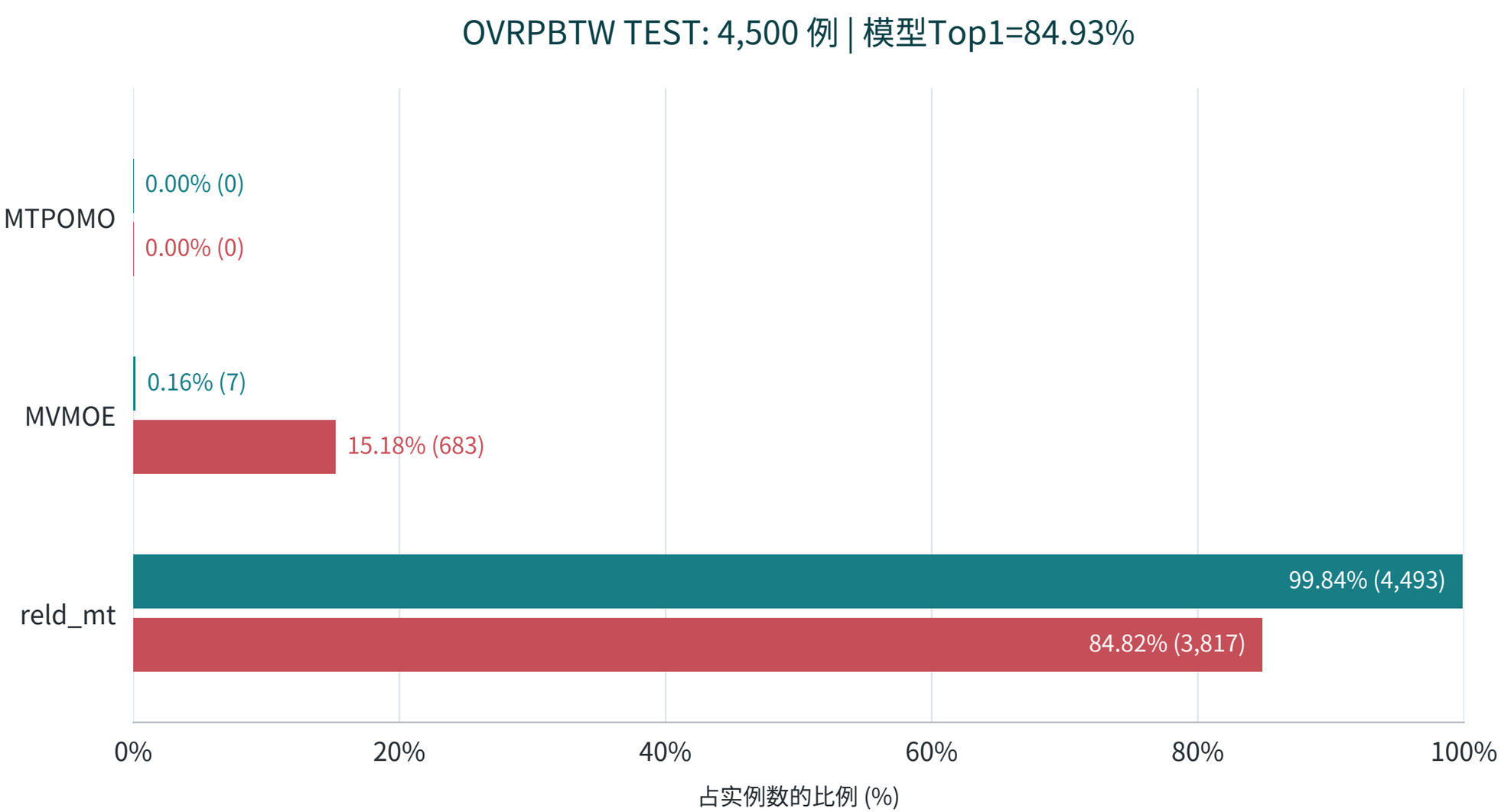
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | OVRPBTW | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

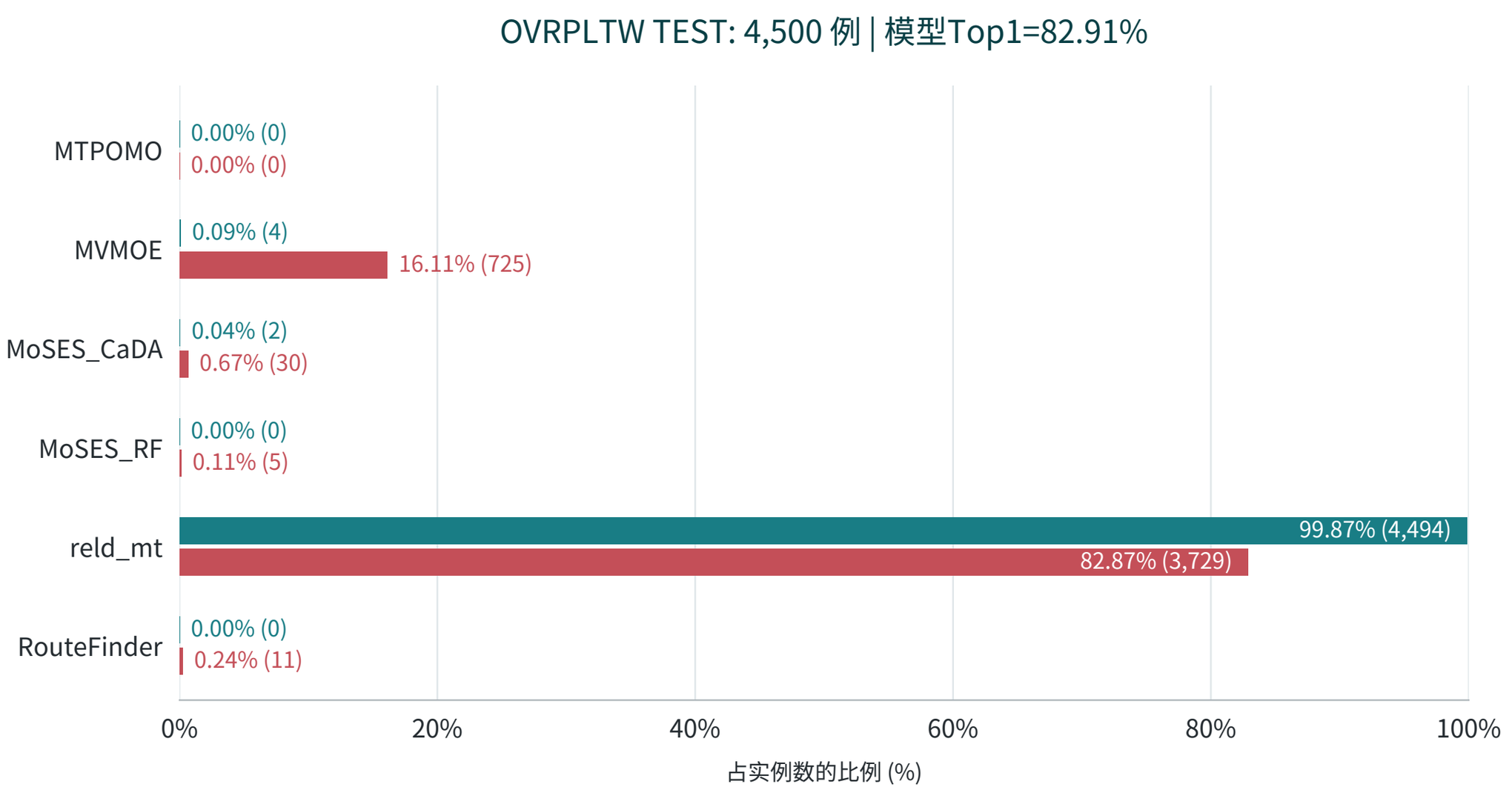
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | OVRPLTW | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

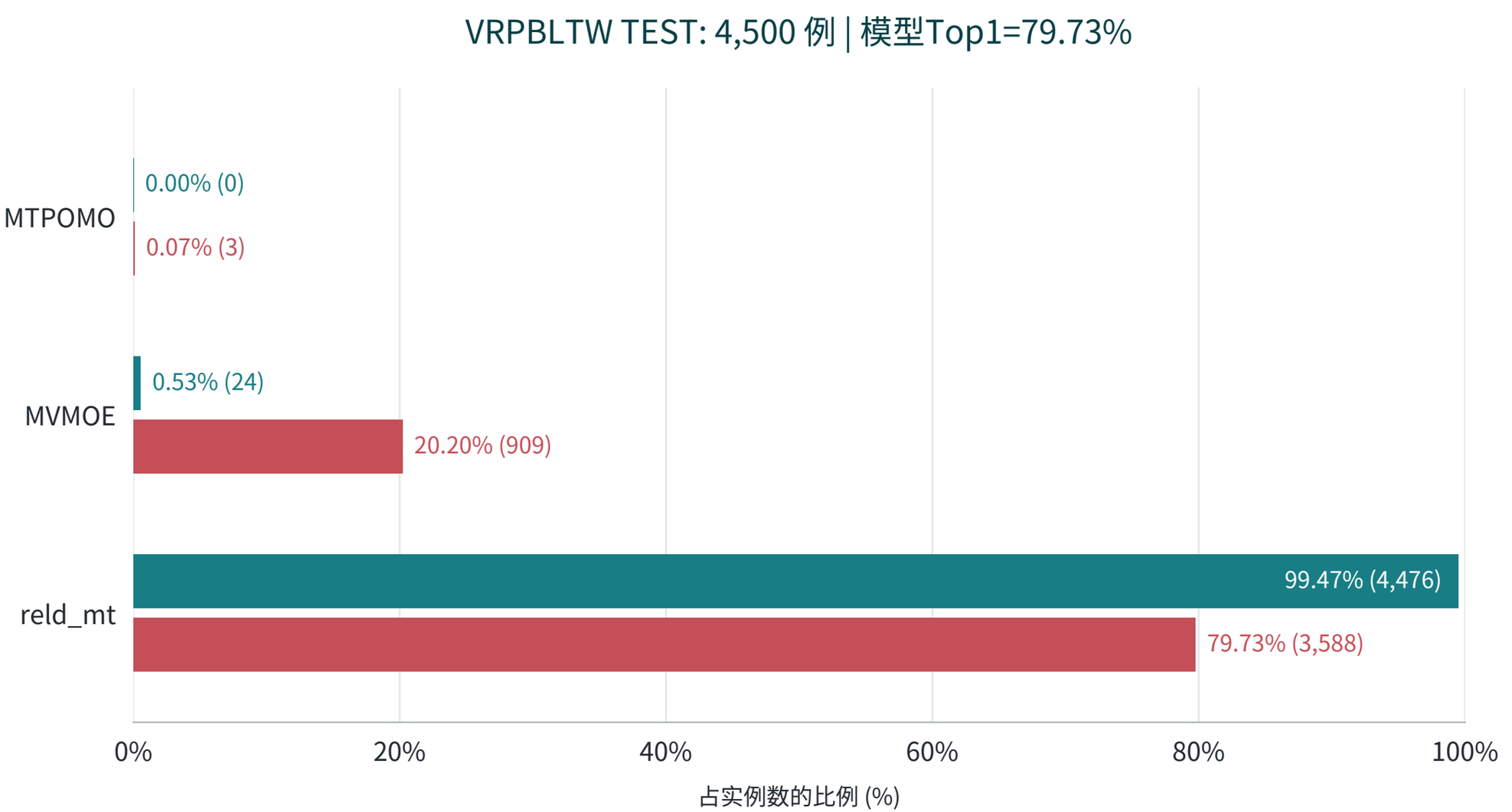
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | VRPBLTW | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

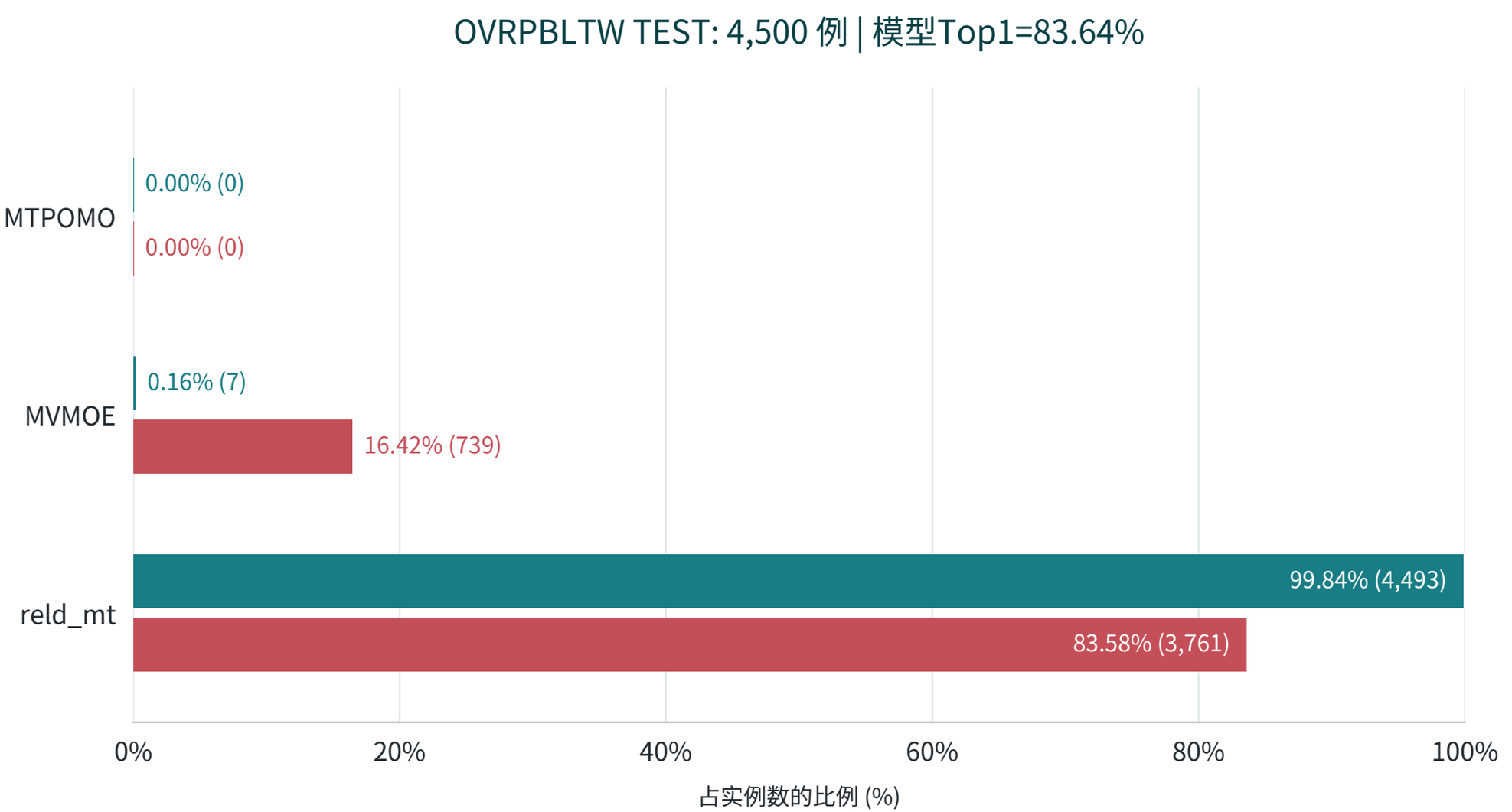
R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/B | OVRPBLTW | 方法分布

同一锁定模型: epoch 34 / update 48960 / SHA256 ccc0ae35e450d184

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

R99/B: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.